StudyOS Thesis-Finder

From vague interests to a prepared first contact — with no database to maintain

StudyOS Team

July 2026

Department of Computer Science, University of Tübingen

The problem, from both sides

Students

- Spend weeks figuring out which chairs even work on their interests
- See 4–5 listed topics, assume that is the whole lab, and never reach out
- Send generic cold emails to the wrong chairs

Professors

- Receive far more inquiries than they can supervise
  (one chair: 50+ inquiries for 20 slots)
- Lose time on rejection emails for mismatched inquiries
- Get low-signal first contacts that miss the chair's actual research

Students and professors both lose when the first contact is poorly matched.

What 27 professors told us — before we built anything

We interviewed 27 professors in the CS department. The verdict on a classic *topic-listing platform* was blunt:

- ∼**50%** do not want a central platform — either already overrun with applicants, or convinced topics must be *co-created in conversation*, not browsed from a list
- ∼**35%** would try a tool — but only if zero-friction, with no reminder emails, and stable for years
- ∼**15%** recruit through other channels entirely

"Profs never update the topics. We've had many attempts in the department already."

A topic platform was tried here around 2022 and failed — for incentive reasons, not UI reasons.

The plan we started with — and why we abandoned it

First attempt

A hosted web app — FastAPI · Celery · Postgres+pgvector · React — scraping professors and serving a match portal.

- Covered **CS Tübingen only**, would have gone stale within months
- Assumed HiWis would keep a scraping pipeline alive "forever", but with low effort
- But still needed someone to maintain the database

The pivot

A database is the wrong asset.

The professors' own signal pointed the way: *build for students, read what already exists, ask chairs for nothing.*

So instead of maintaining data, we packaged *the search process*: a portable agent skill that uses the live web as its source of truth.

The idea: ship the method, not the data

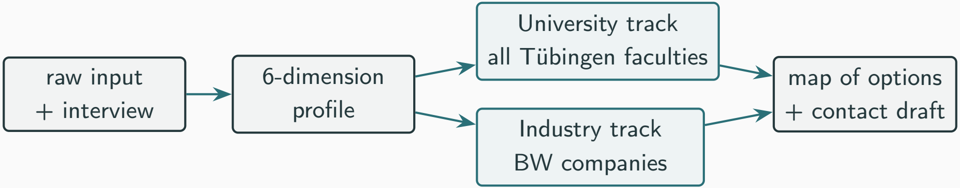
A tool that requires zero professor participation and has no database to keep fresh.

Setup is copy-and-paste: drop the skill folder into any standard AI agent (Claude, ChatGPT, Gemini). No database, no account, no custom app. The student just asks for thesis help, and the agent walks them through three steps:

1. **Formulate** — a guided interview turns vague interests into a structured six-dimension profile
2. **Discover** — live search surfaces which chairs, groups, and companies actually work on relevant problems
3. **Prepare** — draft a specific, high-signal first contact

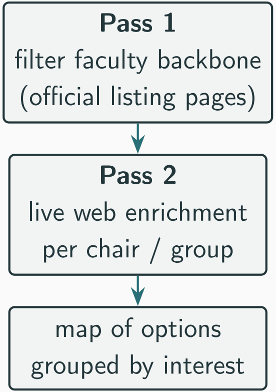
The deliverable is a map of the possibilities — "*now I know what exists and where to start.*"

How it works: one entry point, two tracks



- Input as thin as *"I like deep learning and robotics"* — the agent interviews one question at a time to fill the gaps
- Profile dimensions: *interests* · *methods* · *domain* · *thesis style* · *skills* · *no-go's*
- Every option is grouped by interest, carries a fit rationale and dated evidence, and ends with an honest coverage caveat

The core IP: a search method, not a store



The value over “just ask Claude” lives in two curated references the skill carries:

- **Faculty backbone** — the official `uni-tuebingen.de` listing pages per faculty, crawled first so discovery starts from the real org chart
- **Search strategy** — the reusable mapping from profile dimensions to precise queries, plus filters, dedup, and routing

A parallel ∼100-company BW backbone anchors the industry track. Everything is plain, reviewable Markdown. No runtime database.

Why no database

Database approach (abandoned)	Database-less skill (this project)
Goes stale within months	Always current — reads the live web
CS Tübingen only	All faculties, immediately
Needs maintenance	Zero ongoing maintenance
Only covers what is curated	Covers what is publicly visible
Does not scale well to other faculties	Companies use the same principle

The skill is the intelligence; the data comes from the world.

Privacy and governance

- **Student data never leaves the session.** The profile (interests, coursework, constraints) exists only in the student's own agent conversation — nothing is stored or transmitted by us.
- **Only public academic data is used**: names, roles, research areas, and publications as published by chairs themselves — GDPR-conscious by design.
- **Everything is inspectable.** Skills and references are plain text in a public repository; therefore easily adaptable by hand or AI agent.
- **No fabricated numbers.** A hard project rule: no placeholder metrics, no invented team sizes or citation counts.

Where we are, and what comes next

Behind us

- Talked to professors and understood what actually goes wrong
- Dropped the full-stack app; built the skill-based tool instead
- Made it cover every Tübingen faculty — and companies as a second track

Right now

- Making the tool genuinely *good*, not just working: refining how it searches and improving our evaluation pipeline that measures how well it finds the right chairs

Still to come

- A hands-on test with a handful of real students from different faculties, to see if the tool actually helps them find a thesis
- Getting it into students' hands — starting with the CS Fachschaft and Bachelor's/Master's thesis information session.

What we ask the professors (very little)

1. **Nothing, by default.** The tool works from your public pages and publications as they are.
2. **Optionally:** a few research-area keywords, or a plain "looking for / not looking for" note.
3. **Feedback:** if the inquiries you receive start arriving better-prepared, tell us — that is our success metric.

Students get better discovery. You get fewer but better inquiries.
Nobody is forced to maintain anything.

Thank you — questions?